

Birla Vishvakarma Mahavidyala Engineering College

The Robotics Society, India Student chapter

Technical Report on Faculty Industrial Visit to NAMTECH – AM/NS India and IIT Gandhinagar

1. Objective:

The primary objective of the visit was to provide faculty members from Charutar Vidya Mandal's Birla Vishvakarma Mahavidyalaya (BVM) TRS Student Chapter with exposure to advanced industrial practices and cutting-edge research facilities. This initiative aimed to enhance the faculty's understanding of modern manufacturing technologies and innovative educational methodologies, thereby enriching the academic environment at BVM.

2. Details of the Visit:

Date and Day: 30-01-2025, Thursday

Participants:

- Dr. Mehfuza Holia (Electronics Department)
- Dr. Zankhana Shah (IT Department)
- Prof. Dharita Patel (Electrical Department)
- Prof. Jayesh Koisha (Mechanical Department)
- Dr. Anis Vahora (Electronics and Communication Department)
- Dr. M.M. Solani (Electronics Department)
- Prof. Vijaypratap Singh (Mechanical Department)
- Prof. Rugnesh Vora (JRF, CoE- Smart Manufacturing)

3. Visit to NAMTECH – AM/NS India:

NAMTECH (New Age Makers Institute of Technology) is an educational initiative by ArcelorMittal Nippon Steel India, focusing on Industry 4.0 technologies. The institute offers programs in digital and smart manufacturing, aiming to create a skilled workforce for India's manufacturing sector. The campus features a blend of theoretical and practical training environments, ensuring hands-on learning experiences.

Prof. Dishank Upadhyay and Prof. Keyur Surti from NAMTECH guided the visiting faculty members through all laboratories, providing detailed insights into their functionalities and industrial applications.

Laboratories Explored:

- **Smart Manufacturing Lab:** Equipped with advanced manufacturing setups that integrate digital technologies to optimize production processes, enabling real-time monitoring and control.
- **Industrial Internet of Things (IIoT) Lab:** Demonstrated the integration of smart sensors and cloud-based systems to collect and analyze data for enhanced industrial automation.
- **Robotics Lab:** Featured state-of-the-art robotic arms and autonomous systems used in industrial automation and manufacturing.
- **3D Printing and Additive Manufacturing Lab:** Showcased various 3D printing technologies, including metal and polymer additive manufacturing, with applications in prototyping and production.
- **Cyber-Physical Systems Lab:** Highlighted the integration of computational and physical processes to improve industrial efficiency and automation.

4. Visit to IIT Gandhinagar:

IIT Gandhinagar has established 'Maker Bhavan', a state-of-the-art academic maker space designed to promote active learning, creativity, and innovation among students. Inspired by leading global institutions, the facility includes collaborative teamwork areas, a creative studio, and a variety of digital fabrication tools.

Mr. Ashutosh M from Maker Bhavan, IIT Gandhinagar, guided the visiting faculty members through all laboratories, providing insights into their research, equipment, and applications in innovation and education.

Laboratories Explored:

- **3D Printing Labs:** Provided a comprehensive overview of additive manufacturing techniques, including SLA, SLS, and FDM printing technologies, for research and development.

- **Robotics Lab:** Showcased advanced robotic platforms used for automation, machine learning applications, and human-robot interaction studies.
- **Maker Bhavan:** A multidisciplinary maker space that encourages hands-on learning and fosters collaboration between engineering and design disciplines.
- **Innovation & Creation Lab:** Focused on prototyping, product development, and interdisciplinary research, integrating AI, IoT, and embedded systems.

5. Outcomes:

1. **Enhanced Knowledge:** Faculty members gained deeper insights into the latest advancements in smart manufacturing, IIoT, robotics, and additive manufacturing, reinforcing the importance of modern technologies in academia.
2. **Curriculum Development:** Insights from the visit will support the integration of emerging technologies and methodologies into BVM's academic curriculum, ensuring students are industry-ready.
3. **Research Collaboration:** The visit facilitated discussions on potential research collaborations between BVM and the host institutions, particularly in industrial automation and smart systems.
4. **Industry-Academia Linkage:** Strengthened connections with industry experts and academicians, fostering opportunities for guest lectures, technical workshops, and joint projects.

6. Acknowledgments:

The TRS Student Chapter extends its heartfelt gratitude to Prof. (Dr.) Vinay J. Patel, Head Mechanical Engineering Department and Advisor, TRS student chapter BVM for facilitating this valuable visit for faculty members. Sincere appreciation to Principal BVM, Prof. (Dr.) I. N. Patel, and CVM Chairman, Er. Bhikhubhai Patel, for their continuous support and encouragement. Special appreciation is extended to the Heads of Departments for their guidance and encouragement in making this visit a success.

Special thanks to Dr. Devang Joshi, Registrar, and Prof. Dishank Upadhyay, Prof. Keyur Surti from NAMTECH, and Mr. Ashutosh from Maker Bhavan, IIT Gandhinagar, for their support and coordination during the visit.

7. Glimpses:



